

# IDIR, YACINE

ML / Full-Stack Engineer

Github: [github.com/IDIRYACINE](https://github.com/IDIRYACINE)

PhD in Deep Learning for Medical Image Analysis

Remote — UTC+1 (full CET overlap) — contractor-ready (Deel / Remote)

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## Summary

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PhD-trained deep-learning engineer who also ships production systems end to end. Designed compact medical-imaging networks that match ResNet-18 accuracy at  $\sim 1/100$  of the parameters, while building multi-tenant web platforms, real-time services, and shipped mobile apps. Now leading a small engineering team on an AI developer-tools product.

## Education

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- PhD, Deep Learning for Medical Image Analysis — Mohammed El Bachir Ibrahimi University, Algeria, 2025
- MSc, Business Intelligence — Mohammed El Bachir Ibrahimi University, Algeria, 2024
- BSc, Computer Science — Mohammed El Bachir Ibrahimi University, Algeria, 2022

## Professional Experience

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### Infraxcode | Tech Lead (Hybrid — Algiers / France)

June 2026 – Present

- Lead a team of **3 developers** reporting directly to the **CTO**; own **agile planning**, prioritization, scheduling, and the mentoring and growth of the team.
- Own **product discovery** and **market research**, turning early-stage concepts into shipped products across **6 rapid product-vision iterations** to reach the current direction.
- Set **technical direction** and tooling to accelerate delivery cycles; established onboarding guidelines and a standardized **AI-assisted workflow** that maximizes team productivity with **Claude Code**.
- Build the product hands-on: a **VS Code extension** and **Electron** desktop app powered by the **Claude Agent SDK**.
- Delivered an **MCP server** integration for **business intelligence**, letting AI agents query and analyze business data.
- Co-led delivery on **Bebop** alongside the CTO and a fellow lead, contributing bug fixes, new features, and the **redesign** of the **mobile and web apps**.
- Run internal **feedback analysis** to steer product direction, closing the loop between discovery, build, and iteration.

### Hader | Co-founder & Full-Stack Developer

September 2025 – June 2026

- Improved performance by implementing **caching** through **Tanstack Query** and **optimistic updates** on successful API calls.
- Integrated **Firebase** crash analytics and **push notifications**.
- Architected a complex role-based web app using **Tanstack Router** and **custom role-check hooks**.
- **Dockerized** and deployed an **Odoo** instance for internal enterprise needs, using **Dokploy** on a **Hetzner Cloud VPS**.
- Implemented a scalable **REST API** service layer with **Axios** for better readability and re-usability.
- Built a responsive dashboard using **Tailwind CSS** and **Mantine**.
- Implemented dynamic filters and pagination using **Ag-Grid** and **Tanstack Query**.
- Used hybrid state management — **Zustand** for global state and **Context** for screen-scoped data — to avoid memory degradation on large datasets.
- Developed and shipped 2 **Flutter** apps to the [Play Store](#).

### Ibdapro | Lead Developer

February 2025 – September 2025

- Led planning, design, and delivery of **3 projects**, including a **government-grade** platform.
- Improved **PostgreSQL** performance via **indexes** and **materialized views**.
- Implemented real-time chat using **WebSockets**.
- Migrated a legacy **Node.js / Express.js** codebase to **NestJS**, improving maintainability and scalability.
- Implemented encrypted file reads and uploads using **AES** encryption.
- Designed and implemented a responsive dashboard using **Ant Design** and **Tailwind CSS**.
- Built a **reverse proxy** routing API requests to **region-based instances**, each representing a legally separate entity.
- Implemented **Redis** caching to improve API performance on critical, high-frequency requests.
- Developed and shipped 2 **Flutter** apps to the [Play Store](#).

- Developed a **Java Android** app with 3D rendering using OGRE.
- Built **JNI** wrappers to communicate with custom **native C++ binaries**.
- Completed a computer-vision project pending **U.S. patent (63/199,424)**.
- Developed a **homography-free algorithm** for fiducial markers in **C++** running at **2–10 ms**.

## Research

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### ConvLatticeNet — Compact CNN for Imbalanced Medical Imaging

2025 – 2026

- Designed a **97K-parameter** network for small, class-imbalanced medical datasets that matches or beats **ResNet-18** on **all 6 MedMNIST** tasks with **~1/100** of the parameters.
- Stays accurate where published lightweight networks lose up to **8 AUC points**; validated across **480 controlled runs**.

### DictNet — Data-Efficient Learning Without Pretraining

2025 – 2026

- Built an **82K-parameter** network whose first stage is constructed before training instead of learned, removing the need for **pretrained weights**.
- Beats every from-scratch baseline on every seed across **3 medical tasks**, by **+2.6 to +10.5 AUC**.

### Classifier-Head Ablation for Retinopathy Grading

2025 – 2026

- Ran a **5-factor ablation** across **4 medical tasks**, showing a common classifier-head design misgrades **diabetic retinopathy** further along the severity scale — a harm invisible to **F1** and **AUC**.
- Showed a single linear classifier matches heads **743×** its size.

## Freelance / Consulting

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### Sarl Belmanaa | Odoo Integrator

October 2024 – November 2024

- Delivered an in-house **Odoo ERP** integration, cutting annual operating costs by **~\$38K (USD equiv.)**.

### AnFashion | Consultant & Systems Migration Lead

September 2023 – February 2024

- Resolved a year-long technical conflict between the company and its development agency.
- Managed full infrastructure, reducing monthly running costs by **~60%**.
- Led a complex migration to **Magento 2**, including mail-server recovery and **.dz** domain configuration.